Trainings

Preparatory Steps

With EGR

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

- Drain the cooling system. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/35/35-008-018-tr.html)
- Remove the pressure cap when the engine is cool. Refer to Procedure 008-047 in Section 8. (/qs3/pubsys2/xml/en/procedures/35/35-008-047-tr.html)
- Remove the alternator drive belt. Refer to Procedure 013-005 in Section 13. (/qs3/pubsys2/xml/en/procedures/35/35-013-005-tr.html)
- Remove the alternator. Refer to Procedure 013-001 in Section 13. (/qs3/pubsys2/xml/en/procedures/35/35-013-001-tr.html)
- Remove the alternator mounting bracket. Refer to Procedure 013-003 in Section 13. (/qs3/pubsys2/xml/en/procedures/35/35-013-003-tr.html)
- Remove the coolant supply upper tube. Refer to Procedure 011-031 in Section 11. (/qs3/pubsys2/xml/en/procedures/35/35-011-031-tr.html)

- Remove the thermostat housing. Refer to Procedure 008-015 in section 8. (/qs3/pubsys2/xml/en/procedures/35/35-008-015-tr.html)
- Remove the alternator drive seal. Refer to Procedure 001-001 in Section 1. (/qs3/pubsys2/xml/en/procedures/35/35-001-001-tr.html)

Without EGR

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

- Remove the pressure cap when the engine is cool. Refer to Procedure 008-047 in Section 8. (/qs3/pubsys2/xml/en/procedures/35/35-008-047-tr.html)
- Drain the cooling system. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/35/35-008-018-tr.html)
- Remove the alternator drive belt. Refer to Procedure 013-005 in Section 13. (/qs3/pubsys2/xml/en/procedures/35/35-013-005-tr.html)
- Remove the alternator drive pulley. Refer to Procedure 009-010 in Section 9. (/qs3/pubsys2/xml/en/procedures/35/35-009-010-tr.html)
- Remove the alternator drive seal. Refer to Procedure 001-001 in Section 1. (/qs3/pubsys2/xml/en/procedures/35/35-001-001-tr.html)
- Remove the alternator. Refer to Procedure 013-001 in Section 13. (/qs3/pubsys2/xml/en/procedures/35/35-

013-001-tr.html)

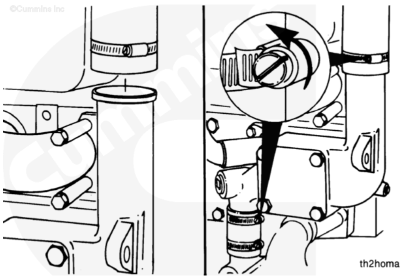
- Remove the thermostat housing support. Refer to Procedure 008-015 in Section 8. (/qs3/pubsys2/xml/en/procedures/35/35-008-015-tr.html)

Remove

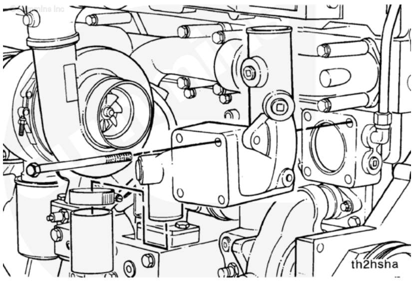
With EGR

Loosen the coolant bypass hose clamps on both the upper and lower hoses.

Remove the upper coolant hose from the thermostat housing.

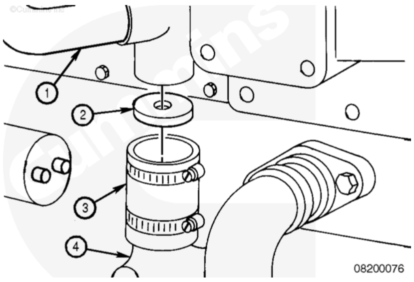


Remove the four thermostat housing mounting capscrews and the thermostat housing.



The coolant flow that provides cooling to the torque converter (if equipped) is achieved in different manners.

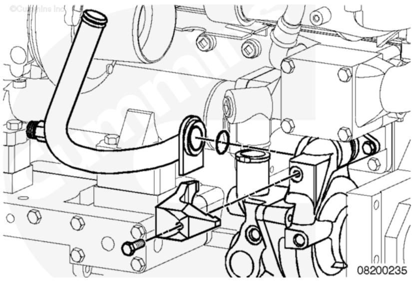
ISM and QSM Series engines use a torque converter cooler disc inside the coolant bypass hose to direct engine coolant to the inlet side of the torque converter cooler.



1. Torque converter coolant supply
2. Torque converter disc (orifice)
3. Bypass hose
4. Water pump.

Loosen the EGR cooler coolant supply lower tube retaining clamp capscrew. Remove the EGR cooler coolant supply lower tube retaining clamp and tube.

Remove and discard the o-ring.

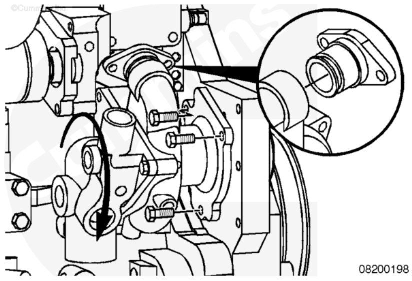


Remove the three water pump mounting capscrews.

Remove the two water pump water transfer connection capscrews and push the water pump transfer connection into the water pump as far as possible to facilitate water pump removal.

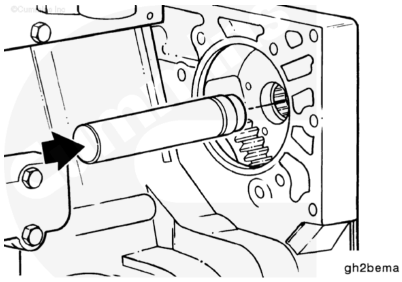
Remove the water pump. Twist the pump outward from the top, and angle the rear of the pump downward, as it is being removed, to allow the pump to pass the thermostat housing support.

Remove the water pump water transfer connection from the water pump.



Check the needle bearing for damage and freedom of needle rotation. Replace the bearing if it is damaged or if it does not turn freely.

If the bearing is replaced, use bearing installation tool, Part Number 3824117, along with a cup driver, to remove the needle bearing from the gear housing. Gently tap the bearing out from the rear side of the housing.

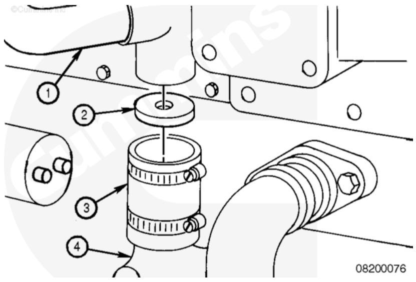


Without EGR

The coolant flow that provides cooling to the torque converter (if equipped) is achieved in different manners.

ISM and QSM Series engines use a torque converter cooler disc inside the coolant bypass hose to direct engine coolant to the inlet side of the torque converter cooler.

1. Torque converter coolant supply
2. Torque converter disc (orifice)
3. Bypass hose
4. Water pump.

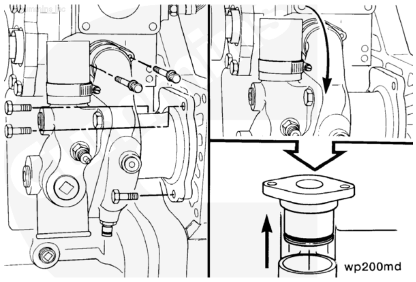


Remove the two water pump water transfer connection capscrews.

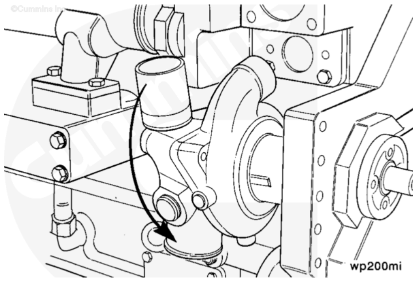
Remove the three water pump mounting capscrews.

Rotate the water pump outward so the water transfer connection can be removed from the water pump.

Remove the water transfer connection from the water pump.

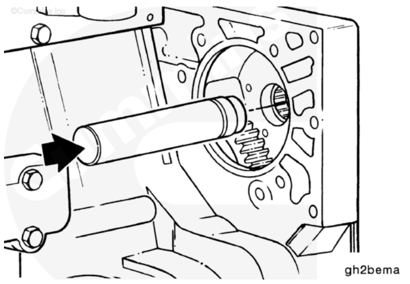


Remove the water pump. Twist the pump outward from the top, and angle the rear of the pump downward, as it is being removed, to allow the pump to pass the thermostat housing support.



Check the needle bearing for damage and freedom of needle rotation. Replace the bearing if it is damaged or if it does **not** turn freely.

If the bearing is replaced, use bearing installation tool, Part Number 3824117, along with a cup driver to remove the needle bearing from the gear housing. Gently tap the bearing out from the rear side of the housing.



Inspect for Reuse

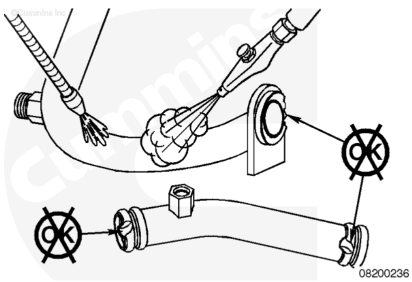
With EGR

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

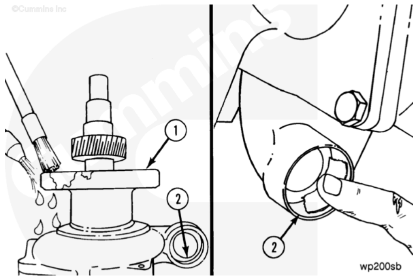


Clean the EGR cooler coolant supply upper and lower tubes with solvent.

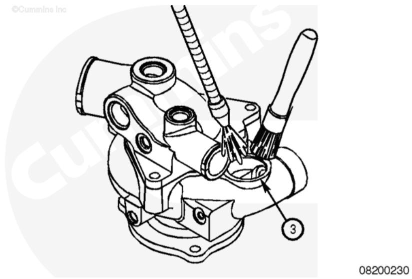
Dry with compressed air.

Inspect the EGR cooler coolant supply upper and lower tubes for cracks or other damage. Replace the component, if it is cracked or otherwise damaged.

Clean surfaces (1 and 2) with solvent and fine grit sandpaper.

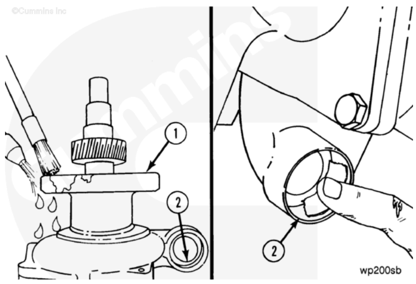


Clean the o-ring (3) sealing surface that interfaces with the EGR cooler coolant supply lower tube.



Without EGR

Clean surfaces (1 and 2).

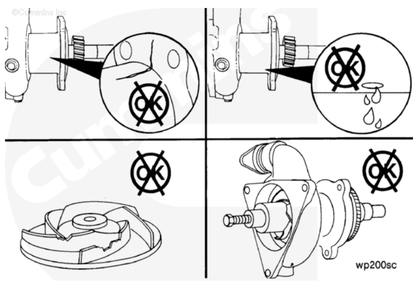


Inspect the water pump housing for cracks or other damage.

Inspect the water pump weep hole for any indication of leaks.

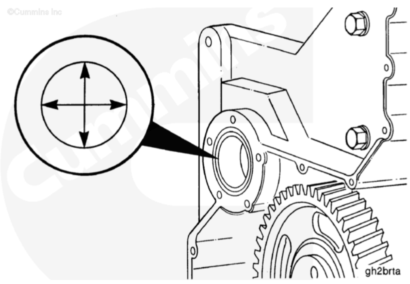
A streak or chemical buildup at the weep hole is **not** justification for water pump replacement. If a steady flow of coolant or oil is observed, replace the water pump with a new or rebuilt unit.

Remove the water pump cover and inspect the water pump impeller for cracks or other damage.



If the needle bearing was removed, measure the inside diameter of the gear housing needle bearing bore.

Needle Bearing Bore Inside Diameter (Gear Housing)		
mm		in
36.967	MIN	1.4554
36.992	MAX	1.4564

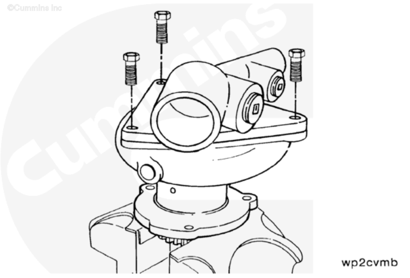


Disassemble

Note : Clamp on the gear portion of the water pump.

Install the water pump in a vise with brass jaws, with the inlet cover facing up.

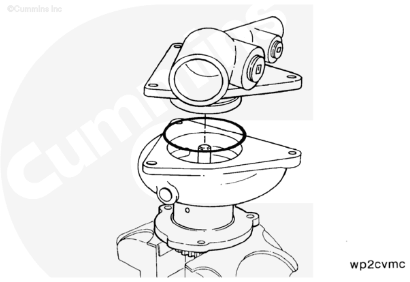
Remove the three capscrews.



Use a mallet to loosen the cover. Remove the inlet cover from the water pump body.

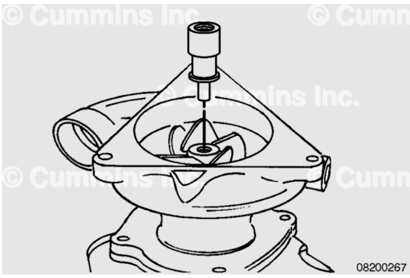
Remove the o-ring from the cover.

Discard the o-ring.

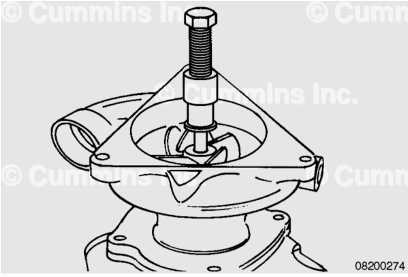


Use the water pump impeller puller kit, Part Number 3376542, or equivalent, to remove the impeller.

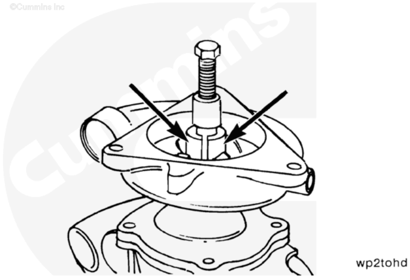
Install the spacer pin into the bore of the water pump impeller.



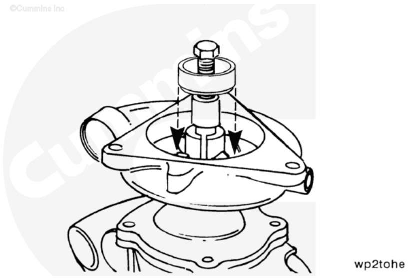
Install the puller sleeve, Part Number 3376544, or equivalent, and the pressure screw, Part Number 3376546, or equivalent, over the spacer pin.



Install the puller halves, Part Number 3376549 or equivalent, on the threaded sleeve assembly.



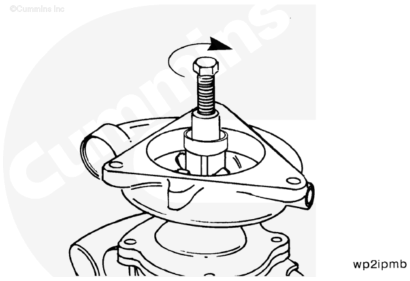
Install the compression sleeve, Part Number 3376545 or equivalent, over the two puller halves.



Hold the puller sleeve with an adjustable wrench.

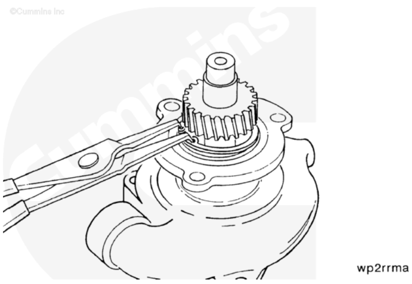
Turn the pressure screw **clockwise** to remove the impeller.

Remove the service tools and the impeller.



⚠ CAUTION ⚠

The retainer ring will be bent or broken if it is removed before the bearing is removed from the shaft.

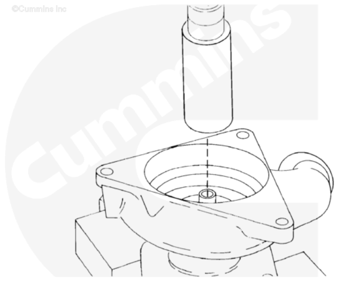


Use a pair of long nose angle snap ring pliers to remove the front bearing retainer ring from the water pump body.

Leave the retainer ring loose on the shaft until the bearing is removed.

Note : The water pump needs to be supported by two 20 cm [8 in] blocks to allow the shaft to drop out of the bottom of the water pump.

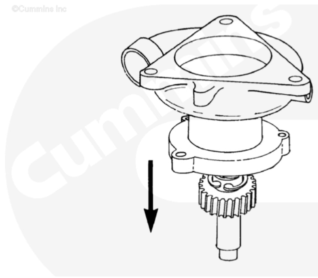
Install the water pump in an arbor press with the front mounting surface facing down.



wp2shma

⚠ CAUTION ⚠

To avoid damage to the water pump shaft, do not allow the shaft assembly to fall when pushed from the water pump body.

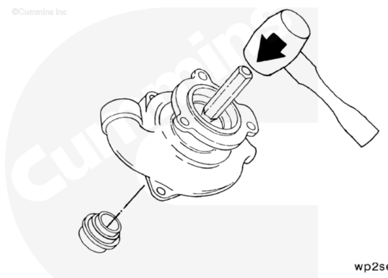


wp200fa

Press the shaft assembly from the water pump body.

⚠ CAUTION ⚠

To prevent a future water pump malfunction, do not damage the water seal bore when removing the water seal.



wp2semg

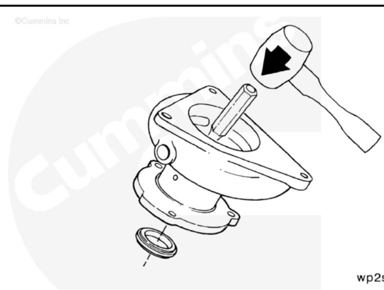
Install the water pump body on a bench with the front mounting surface facing up.

Use a punch and hammer to remove the water seal.

Discard the seal.

⚠ CAUTION ⚠

To prevent future water pump malfunction, do not damage the oil seal bore when removing the oil seal.

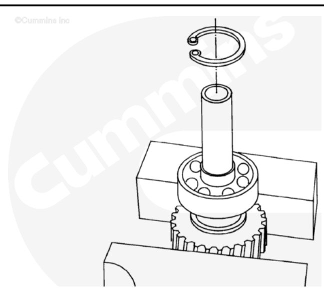


wp2semh

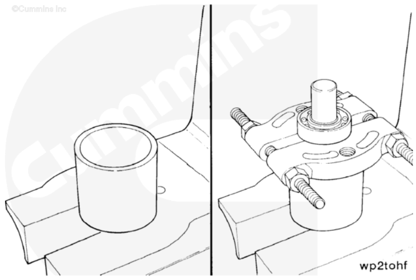
Turn the water pump body over with the front mounting surface facing down.

Use a punch and hammer to remove the oil seal.

Use a pair of long nose angle snap ring pliers to remove the rear bearing retainer ring from the shaft.



wp2rrhb

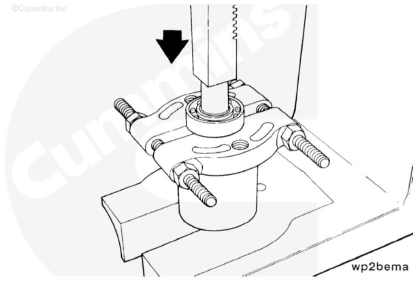


⚠ CAUTION ⚠

The bearing separator halves must be under the inner race of the bearing to prevent personal injury or bearing damage when the shaft is pushed from the bearing.

Install a section of pipe 89 mm [3.50 in] outside diameter x 76.2 mm [3.00 in] inside diameter x 89 mm [3.50 in] length in an arbor press to support the bearing separator.

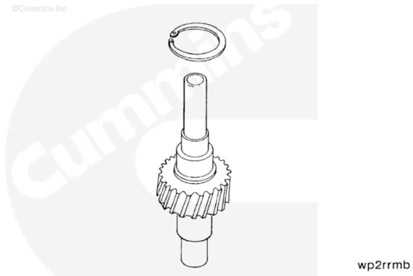
Install the shaft assembly through the section of pipe in the arbor press with the rear bearing inner race supported by bearing separator, Part Number 3375326 or equivalent.



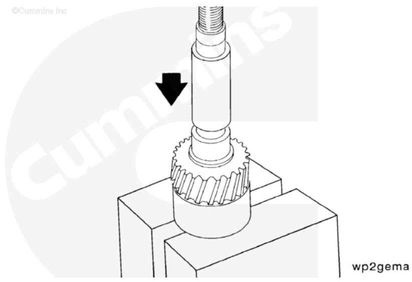
⚠ CAUTION ⚠

To avoid damage to the shaft or personal injury, do not allow the shaft to fall when pressed from the bearing.

Press the shaft from the bearing.



Remove the rear bearing to water pump body retainer ring from the shaft.



⚠ CAUTION ⚠

Do not allow the shaft to fall when pressed from the gear. Damage to the shaft or personal injury can occur.

Note : Make sure the gear surface and **not** the shoulder of the shaft is resting on the arbor press.

Note : To remove the gear from the shaft requires a 13345 N [3000 lbf] arbor press. Do **not** remove the gear from the shaft if there is no damage to the gear.

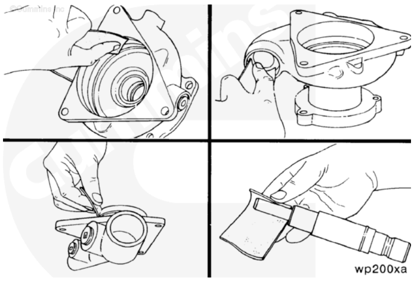
Install the shaft and gear in an arbor press with the largest outside diameter end of the shaft up.

Press the shaft from the gear.

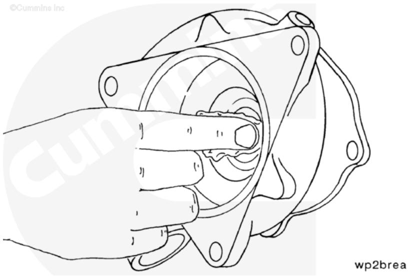
Clean and Inspect for Reuse

Use Scotch-Brite™ 7448 pad, Part Number 3823258 or equivalent, to clean the o-ring bores and grooves in the water pump body and cover.

Use Scotch-Brite™ 7448 pad, Part Number 3823258 or equivalent, to clean the water pump shaft.



Use Scotch-Brite™ 7448 pad, Part Number 3823258 or equivalent, to clean the bearing bores in the water pump body.



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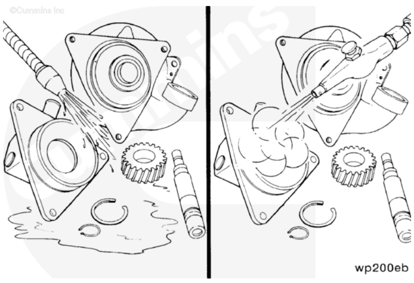


⚠ WARNING ⚠

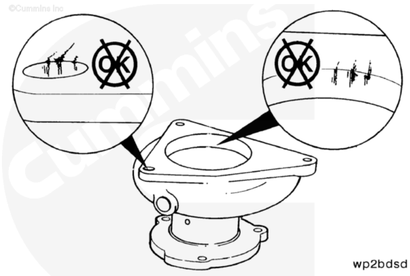
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Use solvent to clean the water pump parts.

Dry with compressed air.

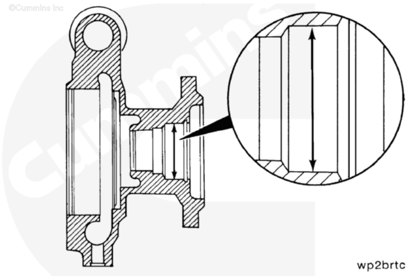
If the part being inspected does **not** meet the specifications given, or is damaged and no alternative is given, the part **must** be replaced.

Inspect the water pump body for cracks, porosity, and excessive corrosion.



Measure the water pump body bearing bore inside diameter.

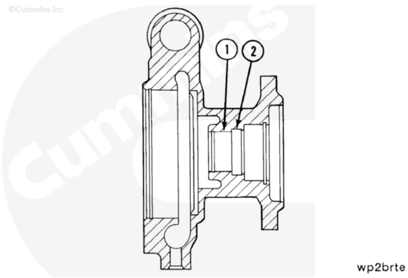
Water Pump Body Bearing Bore I.D.		
mm		in
51.996	MIN	2.0471
52.011	MAX	2.0477



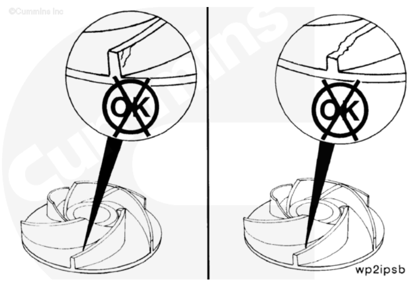
Measure the water seal bore inside diameter (1).

Measure the oil seal bore inside diameter (2).

Seal Bore I.D.			
	mm		in
1	36.450	MIN	1.4350
	36.475	MAX	1.4360
2	40.975	MIN	1.6132
	41.025	MAX	1.6152

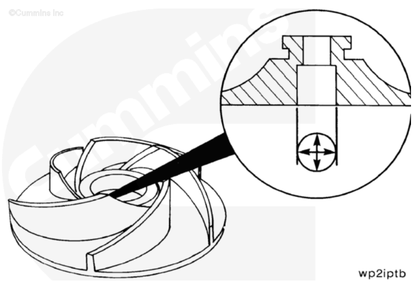


Inspect the water pump impeller for cracks or other damage.

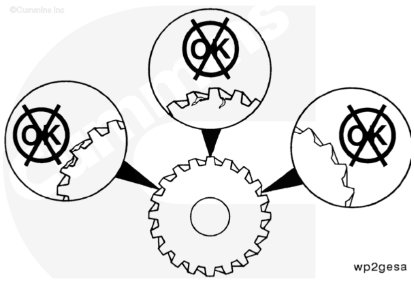


Measure the water pump impeller bore inside diameter.

Impeller Bore I.D.		
mm		in
15.849	MIN	0.6240
15.875	MAX	0.6250

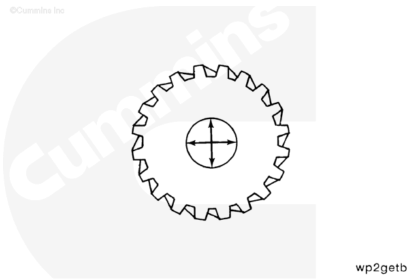


Inspect the water pump drive gear for broken teeth, cracks, and excessive or uneven tooth wear.

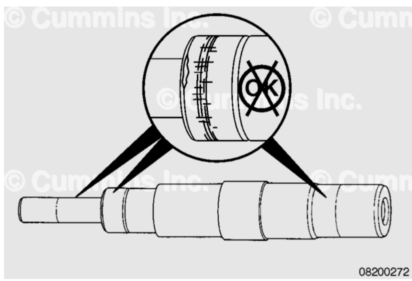


Measure the water pump drive gear bore inside diameter.

Drive Gear I.D.		
mm		in
33.900	MIN	1.3346
33.925	MAX	1.3356

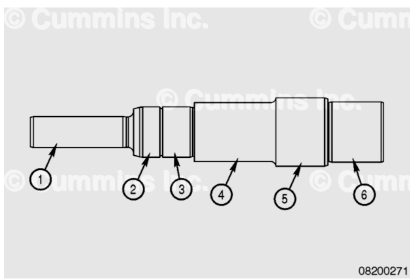


Inspect the seal surfaces of the water pump shaft for large grooves, nicks, or other damage.

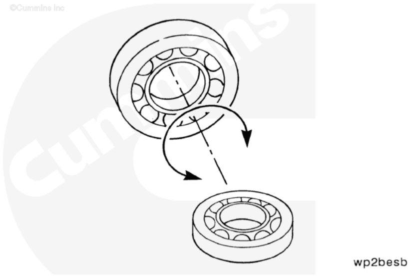


Measure the water pump shaft journal outside diameters.

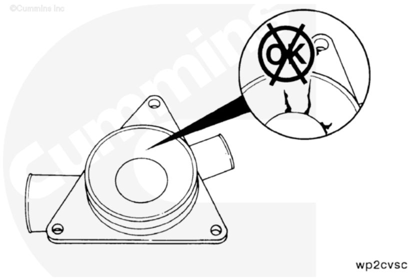
Shaft Journal O.Ds.			
	mm		in
1	15.906	MIN	0.6262
	15.918	MAX	0.6267
2	24.999	MIN	0.9842
	25.009	MAX	0.9846
3	24.999	MIN	0.9842
	25.009	MAX	0.9846
4	29.250	MIN	1.1515
	29.751	MAX	1.1713
5	33.951	MIN	1.3376
	33.976	MAX	1.3367
6	29.987	MIN	1.1806
	30.000	MAX	1.1811



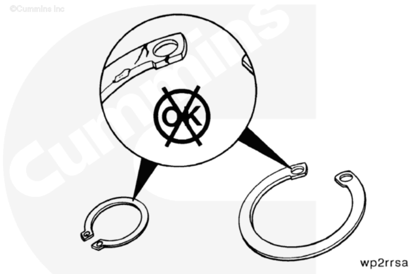
Turn the water pump bearing by hand to inspect for freedom of rotation.



Inspect the water inlet cover for cracks and/or excessive corrosion.

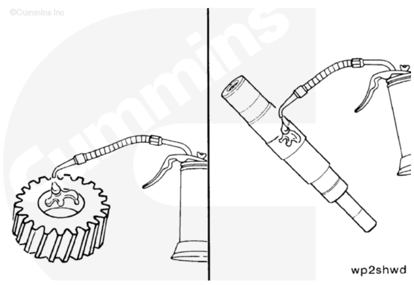


Inspect the bearing retainer rings for nicks and cracks.



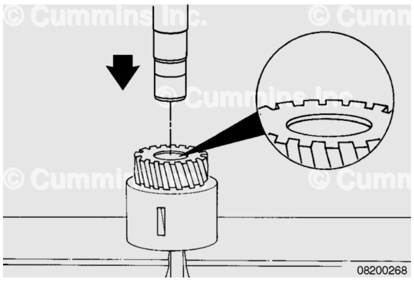
Assemble

Use clean 15W-40 lubricating oil to lubricate the drive gear bore and the shaft.



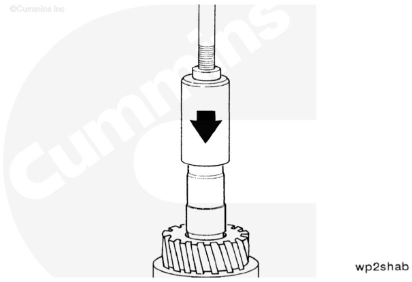
Install the gear in an arbor press with the chamfered side of the gear bore facing up.

Insert the large diameter of the shaft through the top of the gear bore.



Note : Press the shaft through until the tool touches the gear surface.

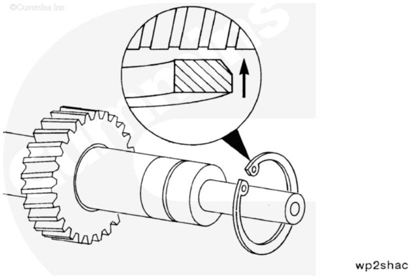
Use the water pump gear mandrel, Part Number 3823851 or equivalent, to press the shaft into the gear.



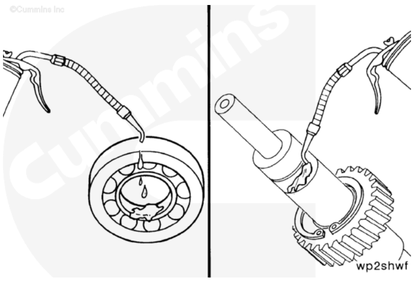
⚠ CAUTION ⚠

To prevent future water pump malfunctions, the beveled side of the retainer ring must be installed toward the drive.

Loosely install the front bearing to water pump body retainer ring on the shaft.



Use clean 15W-40 oil to lubricate the bearing and shaft bearing journal.

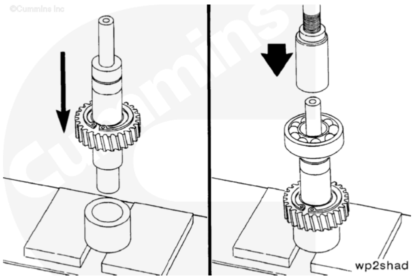


⚠ CAUTION ⚠

Press against the inner race of the bearing to prevent personal injury or bearing damage.

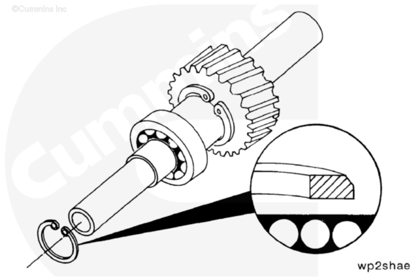
Install the shaft assembly in an arbor press. Use the water pump bearing mandrel, Part Number 3375318, or equivalent, to support the shaft assembly.

Use the water pump bearing mandrel, Part Number ST-658 or equivalent, to press the bearing on the shaft until the bearing touches the shoulder on the shaft.



The flat side of the retainer ring **must** be installed toward the bearing.

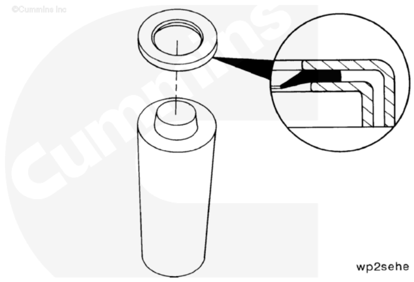
Use a pair of snap ring pliers to install the bearing retainer ring into the groove on the shaft.



Note : Do not lubricate the oil seal. The oil seal must be installed with the lip of the seal and the shaft clean and dry, to provide a proper sealing surface and prevent future water pump and engine damage.

The sealing lip of the oil seal **must** be installed toward the seal driver.

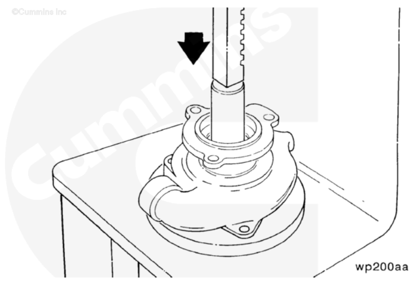
Install the oil seal on the water pump oil seal driver, Part Number 3376139 or equivalent.



Install the water pump body in an arbor press with the oil seal bore (water pump mounting surface) facing up.

Install the oil seal and oil seal driver into the oil seal bore.

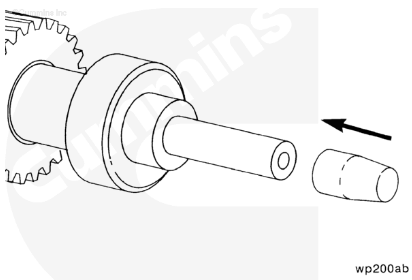
Press the oil seal into the oil seal bore until the seal driver touches the water pump body.



⚠ CAUTION ⚠

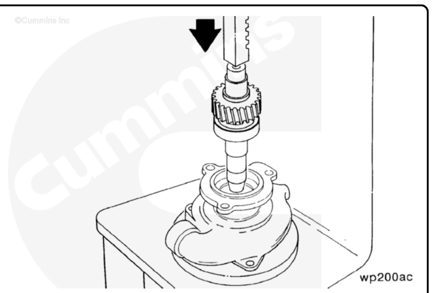
To avoid deformation of the seal lip and water pump leakage, water pump oil seal expander, Part Number 3376387, or equivalent, must be used when installing the shaft assembly.

Install the water pump oil seal expander, Part Number 3376387 or equivalent, on the impeller end of the shaft.



Note : The shaft assembly must be pressed into the body far enough to install the rear bearing retainer ring into the groove of the body.

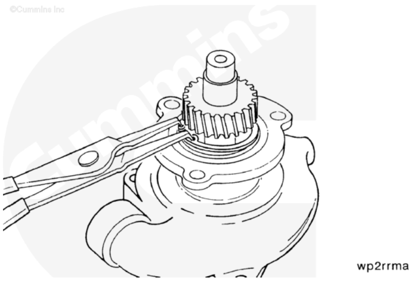
Install the water pump body in an arbor press with the front mounting surface facing up.



Press the shaft assembly and oil seal expander into the water pump body.

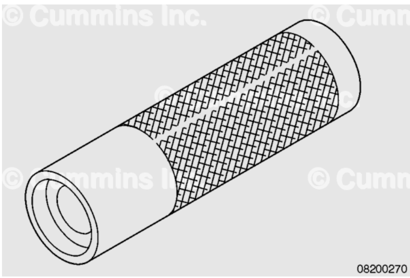
Remove the oil seal expander.

Use a pair of long nose angle snap ring pliers to install the front bearing retainer ring into the groove of the water pump body.



⚠ CAUTION ⚠

The use of any tool other than what is specified can result is seal damge.

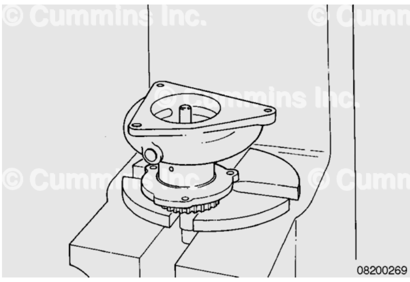


Use the water pump seal driver to press the seal into the water pump body.

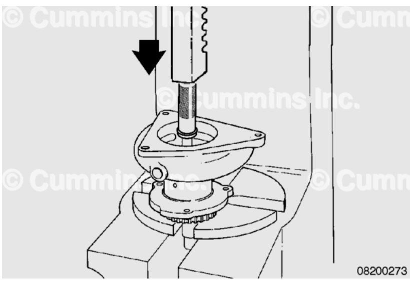
Seal Driver, Part Number 3823815, or equivalent, is to be used for Chicago-Allis™ water seals.

Seal Driver, Part Number 4918797, or equivalent, is to be used for Kaco™ water seals.

Use service tool, Part Number 3823815, Part Number 4918797, or equivalent, to install the water seal into the service tool. Place the service tool on the arbor press table and install the water seal into the service tool. Place the water pump assembly on top of the tool with the small end of the shaft inserted in the seal. Use the arbor press to press the pump into the seal.

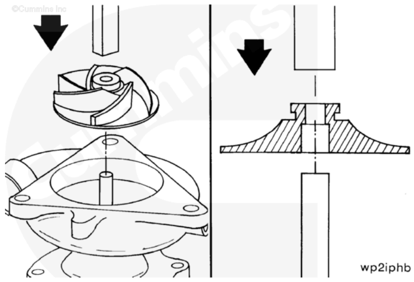


Press the pump until hitting the stop. The seal flange **must** be tight against the pump body.



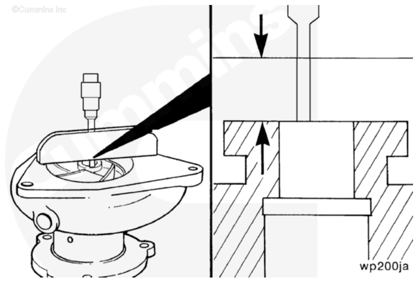
⚠ CAUTION ⚠

Make sure the arbor press shaft is not to large of a diameter. It should not touch the plastic portion of the impeller or damage to the impeller will occur.



Install the water pump body in an arbor press with the impeller end of the shaft facing up. Support the bottom end of the shaft.

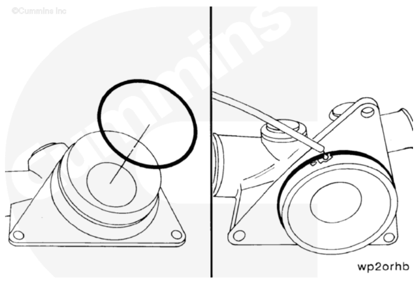
Press the impeller on the shaft until the step inside the impeller bore touches the shaft.



Use a dial depth gauge to measure the distance from the impeller hub to the water pump body surface.

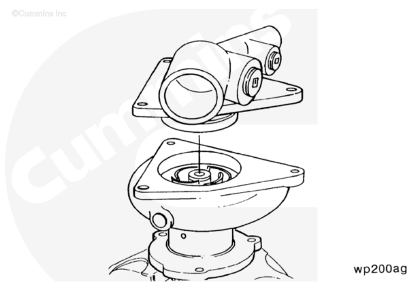
Impeller Hub to Body Surface		
mm		in
13.52	MIN	0.532
13.72	MAX	0.540

If the distance from the impeller hub to the body surface is greater than the maximum specification, use the water pump impeller puller kit, Part Number 3376542 or equivalent, to move the impeller to the proper distance.



Install the new o-ring on the water inlet cover.

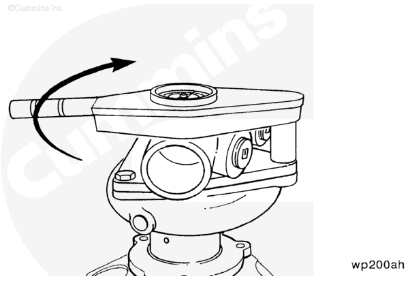
Use clean vegetable oil to lubricate the o-ring.



Install the water inlet cover on the water pump body.

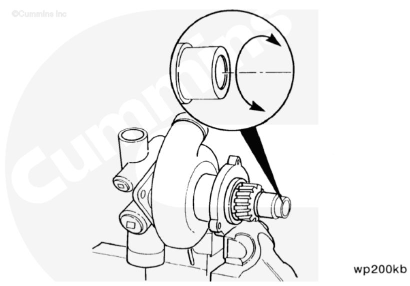
Install three (M10-1.50 x 25) capscrews in the water pump body.

Torque Value: 47 n•m [35 ft-lb]



Turn the water pump shaft by hand to inspect the bearings and impeller for freedom of rotation.

If the water pump shaft does **not** rotate freely in the water pump body, the water pump **must** be disassembled, inspected, and assembled again.



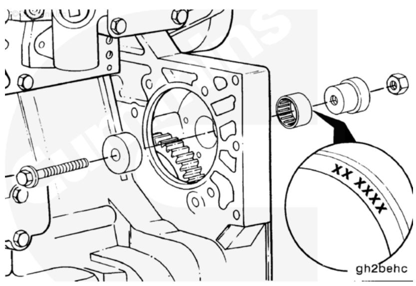
Install

With EGR

If the needle bearing was removed, use bearing installation tool, Part Number 3824117, to install a new needle bearing in the gear housing.

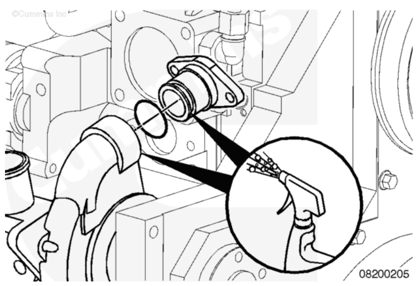
The bearing **must** be installed with the part number side of the bearing against the installation tool to prevent damage to the bearing during the installation.

Install the bearing from the front side of the gear housing until the bearing is flush with the front edge of the housing bore.



⚠ CAUTION ⚠

In order to reduce the possibility of pinching or rolling the o-rings during assembly do not lubricate the o-rings. Lubricate the o-ring connection mating surface with clean engine coolant, soapy water, or vegetable oil.



Install the water pump cover. Use a new o-ring.

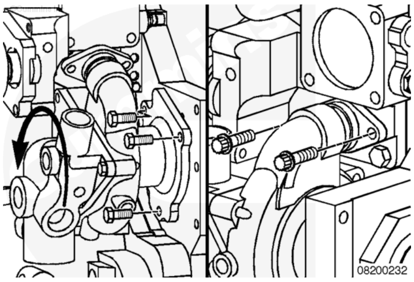
Torque Value: 47 n•m [35 ft-lb]

Install the water pump water transfer connection into the water pump.

Install a new o-ring on the water pump mounting flange.

The water pump **must** be twisted outward from the top until the transfer outlet clears the thermostat housing support during installation.

Install the water pump.

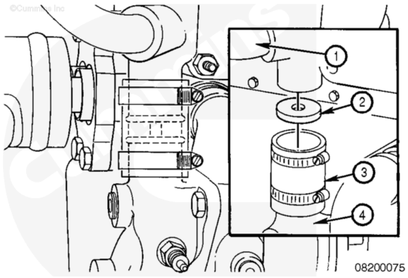


Twist the water pump inward and install the three water pump mounting capscrews.

Torque Value: 47 n•m [35 ft-lb]

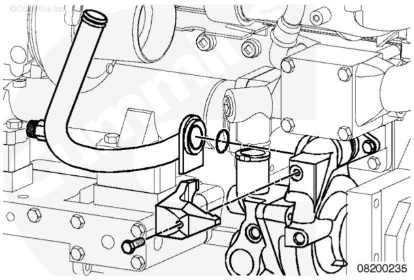
Install a new gasket on the water pump water transfer connection. Install and tighten the water transfer connection capscrews.

Torque Value: 25 n•m [221 in-lb]



If the engine is equipped with a torque converter cooler, install the disc in the bypass hose before installing the thermostat housing.

1. Torque converter coolant supply
2. Torque converter disc (orifice)
3. Bypass hose
4. Water pump.



⚠ CAUTION ⚠

In order to reduce the possibility of pinching or rolling the o-rings during assembly do not lubricate the o-rings. Lubricate the o-ring connection mating surface with clean engine coolant, soapy water, or vegetable oil.

Install a new o-ring on the EGR cooler coolant supply lower tube.

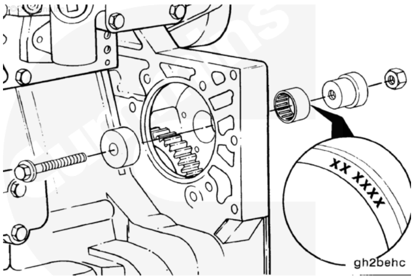
Insert the EGR cooler coolant supply lower tube into the water pump body.

Make sure the EGR cooler coolant supply lower tube is seated completely in the water pump body before tightening the EGR cooler coolant supply lower tube retaining clamp capscrew, to reduce the possibility of pinching the o-ring.

Tighten the capscrew.

Torque Value: 34 n•m [25 ft-lb]

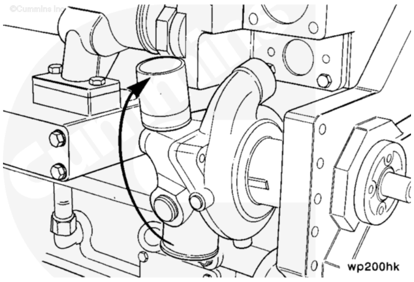
Without EGR



If the needle bearing was removed, use bearing installation tool, Part Number 3824117, to install a new needle bearing in the gear housing.

The bearing **must** be installed with the part number side of the bearing against the installation tool to prevent damage to the bearing during the installation.

Install the bearing from the front side of the gear housing until the bearing is flush with the front edge of the housing bore.



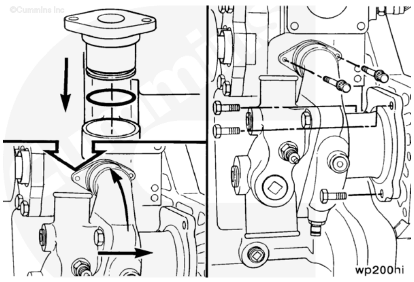
Install the water pump cover. Use a new o-ring.

Torque Value: 47 n•m [35 ft-lb]

Install a new o-ring on the water pump mounting flange.

The water pump **must** be twisted outward from the top until the transfer outlet clears the thermostat housing support during installation.

Install the water pump.



⚠ CAUTION ⚠

In order to reduce the possibility of pinching or rolling the o-rings during assembly do not lubricate the o-rings. Lubricate the o-ring connection mating surface with clean engine coolant, soapy water, or vegetable oil.

Install a new o-ring on the water transfer tube.

Install the connection into the water pump.

Twist the water pump inward and install the three water pump mounting capscrews.

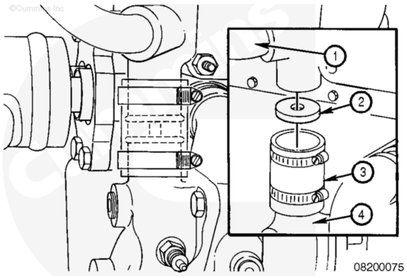
Torque Value: 47 n•m [35 ft-lb]

Install a new gasket on the water pump water transfer connection. Install and tighten the water transfer connection capscrews.

Torque Value: 25 n•m [221 in-lb]

If the engine is equipped with a torque converter cooler, install the disc in the bypass hose before installing the thermostat housing.

1. Torque converter coolant supply
2. Torque converter disc (orifice)
3. Bypass hose
4. Water pump.



Finishing Steps

With EGR

⚠ CAUTION ⚠

In order to reduce the possibility of pinching or rolling the o-rings during assembly, do not lubricate the o-rings. Lubricate the o-ring connection mating surface with clean engine coolant, soapy water, or vegetable oil.

- Install the alternator drive seal. Refer to Procedure 001-001 in Section 1. (/qs3/pubsys2/xml/en/procedures/35/35-001-001-tr.html)
- Install the coolant thermostat housing support. Refer to Procedure 008-015 in Section 8. (/qs3/pubsys2/xml/en/procedures/35/35-008-015-tr.html)
- Install the EGR cooler coolant supply upper tube. Refer to Procedure 011-031 in Section 11. (/qs3/pubsys2/xml/en/procedures/35/35-011-031-tr.html)
- Install the alternator bracket. Refer to Procedure 013-003 in Section 13. (/qs3/pubsys2/xml/en/procedures/35/35-013-003-tr.html)
- Install the alternator. Refer to Procedure 013-001 in Section 13. (/qs3/pubsys2/xml/en/procedures/35/35-013-001-tr.html)
- Install the alternator drive belt. Refer to Procedure 013-005 in Section 13. (/qs3/pubsys2/xml/en/procedures/35/35-013-005-tr.html)
- Install the cooling system pressure cap. Refer to Procedure 008-047 in Section 8.

(/qs3/pubsys2/xml/en/procedures/35/35-008-047-tr.html)
- Fill the cooling system. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/35/35-008-018-tr.html)
- Operate the engine until it reaches a temperature of 82°C [180°F] and check for leaks.

Without EGR

- Install the alternator drive seal. Refer to Procedure 001-001 in Section 1. (/qs3/pubsys2/xml/en/procedures/35/35-001-001-tr.html)
- Install the alternator drive pulley. Refer to Procedure 009-010 in Section 9. (/qs3/pubsys2/xml/en/procedures/35/35-009-010-tr.html)
- Install the coolant thermostat housing support. Refer to Procedure 008-015 in Section 8. (/qs3/pubsys2/xml/en/procedures/35/35-008-015-tr.html)
- Install the alternator. Refer to Procedure 013-001 in Section 13. (/qs3/pubsys2/xml/en/procedures/35/35-013-001-tr.html)
- Install and adjust the alternator drive belt. Refer to Procedure 013-005 in Section 13. (/qs3/pubsys2/xml/en/procedures/35/35-013-005-tr.html)
- Fill the cooling system. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/35/35-008-018-tr.html)
- Operate the engine until it reaches a temperature of 82°C [180°F] and check for leaks.

Last Modified: 03-Jan-2012